

On Default Correlation in Portfolio Modeling

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Default Correlation

- What is the default correlation?
- Traditional Correlation defined in the current finance literature

$$(A,B) = \frac{\Pr [A \cap B] - P[A] \Pr[B]}{\sqrt{P(A)[1 - P(A)]P(B)[1 - P(B)]}}$$

Problems with This Approach

- One year is an arbitrary choice, useful information about the term structure of default rates could be lost
- Default correlation is a time dependent variable
- Need correlation over a number of years instead of only one year
- Estimation of default correlation has its problem

Some Alternative Approaches

- Credit Risk +:
 - Using conditioning on some hidden parameters. It might be useful for large portfolio management, but not for a credit portfolio of the typical size of FTD or CDOs. It might be too
- CreditMetrics and KMV
 - It uses implicitly a normal copula function as what we use in our framework
- CreditPortfolio View
 - It uses macroeconomic data to create links between credits. It requires empirical studies as well as credit classification. It does not give enough details.

Default Correlation: The Joy of Copulas

- We first know the marginal distribution of survival time for each credit
- We need to construct a joint distribution with given marginals and a correlation structures
- Copula function used in multivariate statistics can be used
- The correlation parameters used in copula function can be interpreted as the asset correlation between two credits used in CreditMetrics

What is a Copula Function?

- Function that join or couple multivariate distribution functions to their one-dimensional marginal distribution functions
- For m uniform r. v., $U_1, U_2, \dots, U_m$

$$C(u_1, u_2, \dots, u_m) = \Pr[U_1 \leq u_1, U_2 \leq u_2, \dots, U_m \leq u_m]$$

- Suppose we have m marginal distributions with distribution function $F_i(x_i)$
- Then the following defines a multivariate distribution function

$$F(x_1, x_2, \dots, x_m) = C(F_1(x_1), F_2(x_2), \dots, F_m(x_m))$$

How do we simulate the default time?

- Simulate a joint normal distribution Y_i with a given correlation matrix Σ
- Translate Y into a uniform random variable Z
- Use each credit curve to get survival time for each credit

$$T_i = F_i^{-1}(\Phi(Y_i))$$

Summary of the Simulation



- Use asset approach to default correlation
- Simulate correlated multivariate normal distribution with the asset correlation
- Translate the multivariate normal random variable into survival times by using marginal term structure of default rates

Correlation Engine: Economic Model

- Choose a world wide equity indices
- Map each obligor into countries, industries and sectors
- Calculate asset correlation based on the historical data of equity indices
- Incorporate company size into the calculation of the idiosyncratic risk
- Two obligors are correlated mainly due their sharing of common countries, industries and sectors
- This engine has been implemented in DPL and would like to get feedback from business

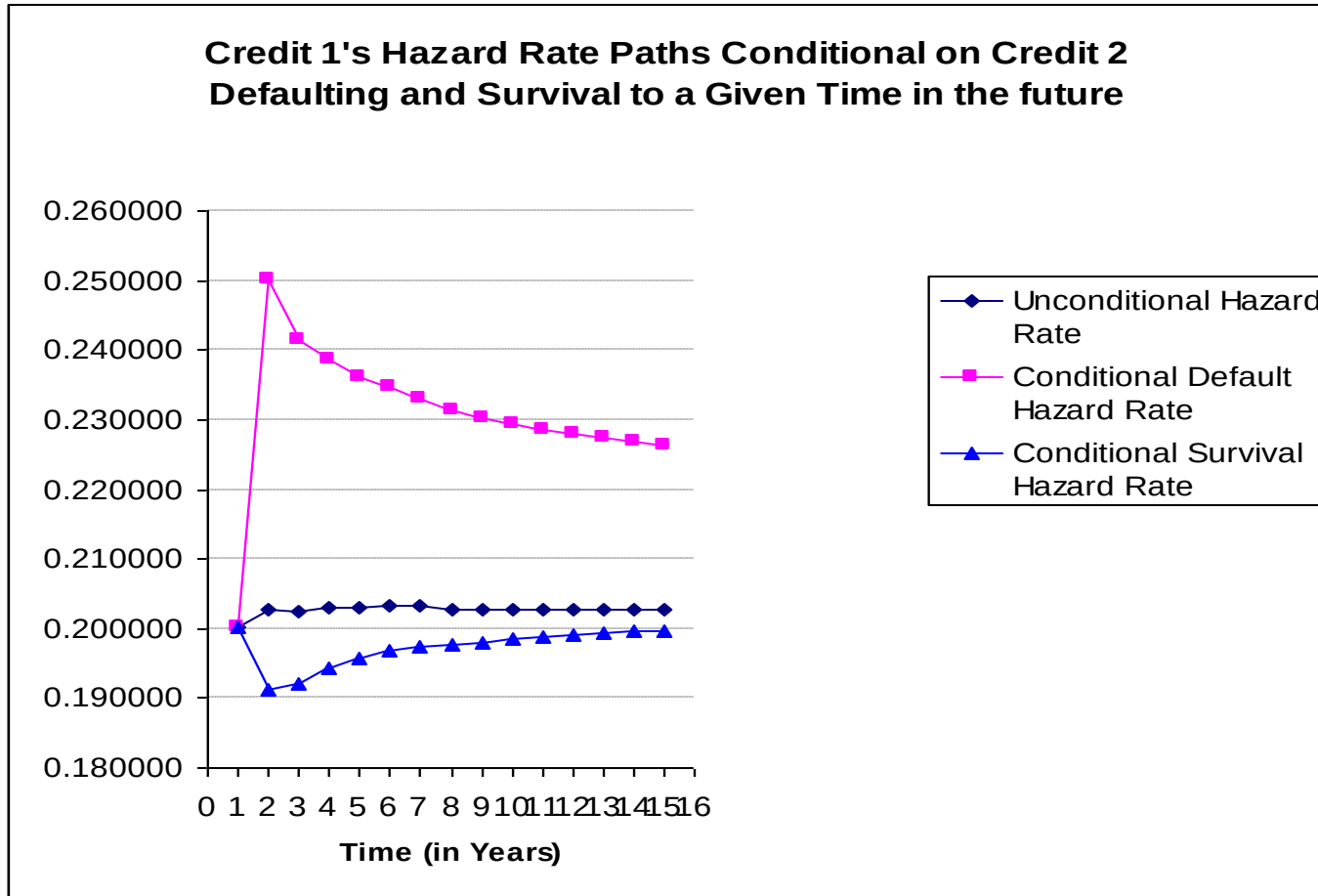
Efficient Credit Portfolio Simulation

- Importance Sampling
 - For single name we should shift the normal mean from 0 to 0.865
- Quasi Monte Carlo
 - It does not work very well for high dimension; but it would help tremendously if we use it in conjunction with the reduction of dimension
- Reduction of Dimensionalities
 - One correlation – one factor model
 - Inter and intra industry correlation – $(n + 1)$ factor model where n is the number of industry groups
- Fast Fourier Transformation Approach (FFT)

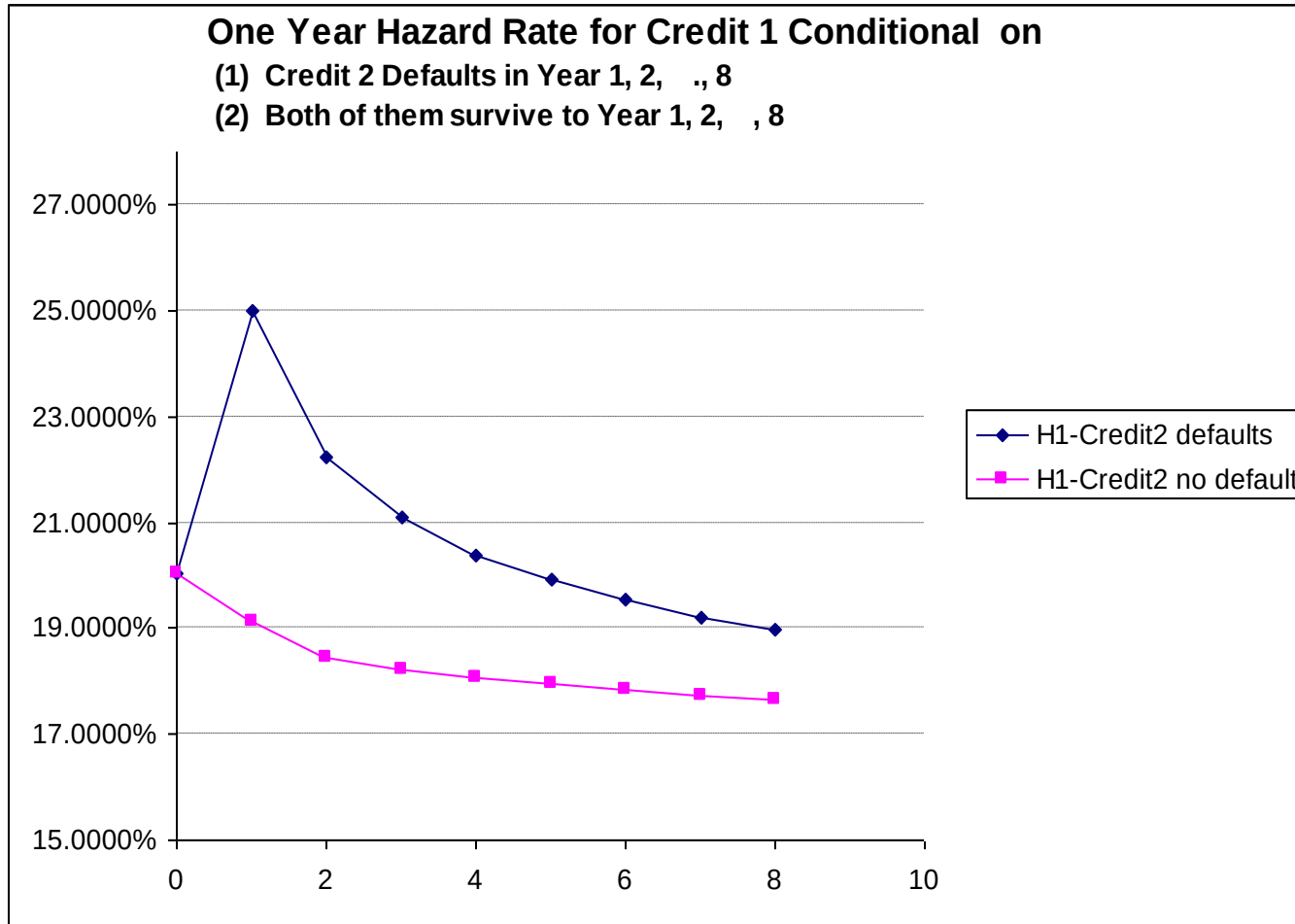
Factor Copula Approach and Conditional Independence

- Introduce latent factors such that the default times are independent conditionally on the factors
- Conditional on these factors, estimate conditional default probabilities
- Joint survival function is a product of individual survival probabilities (it can be expressed analytically for factor copulas)
- One parameter Gaussian copula corresponds to one-factor model of asset returns,
- Two-parameter Gaussian copula (with intra/inter-industry parameters) can be expressed in terms of n industry factors and one market factor

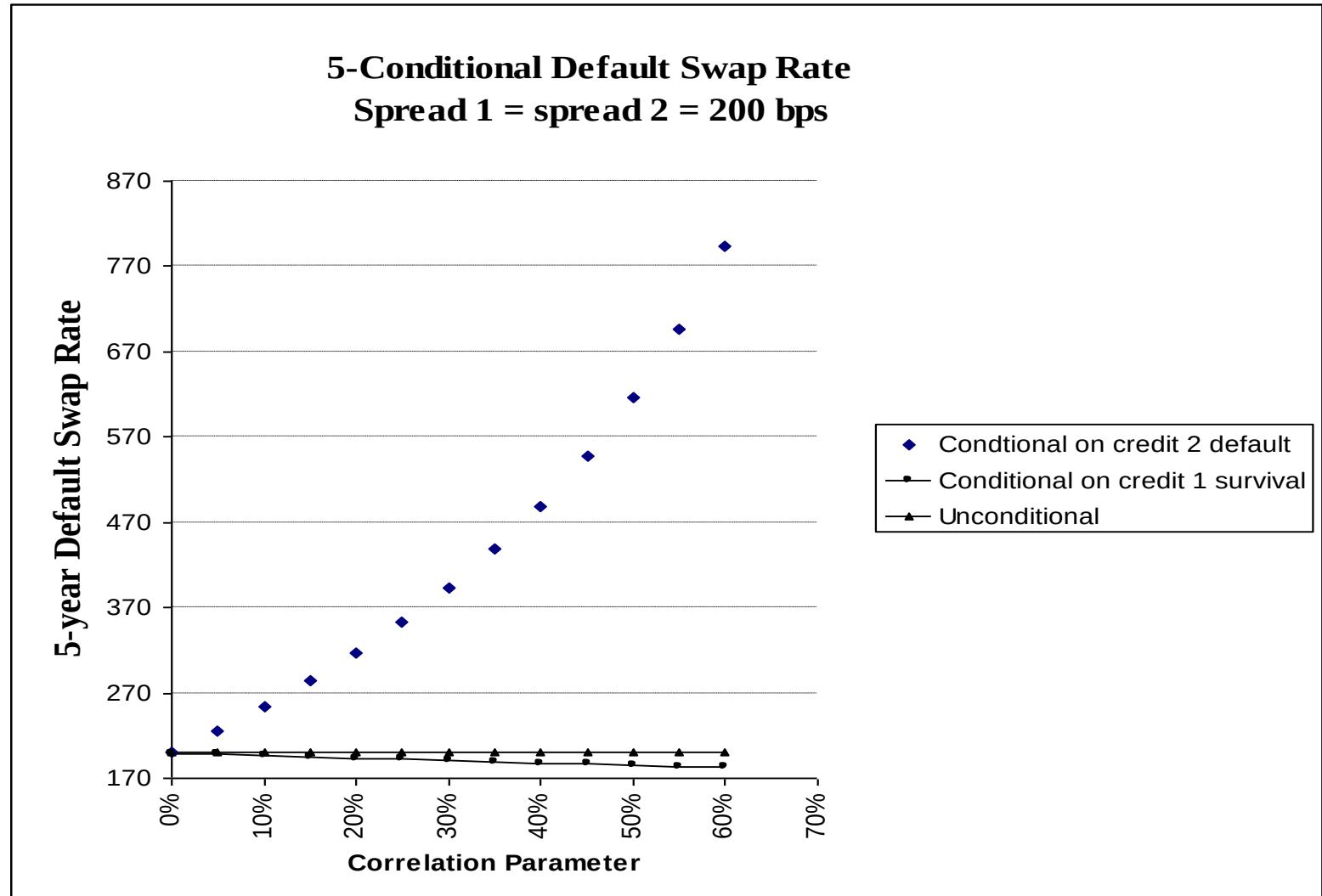
Correlation from Conditional Spread Movement (20% correlation)



Correlation from Conditional Spread Movement (20% correlation, credit spreads = 1000)



Conditional Par Default Swap Rate



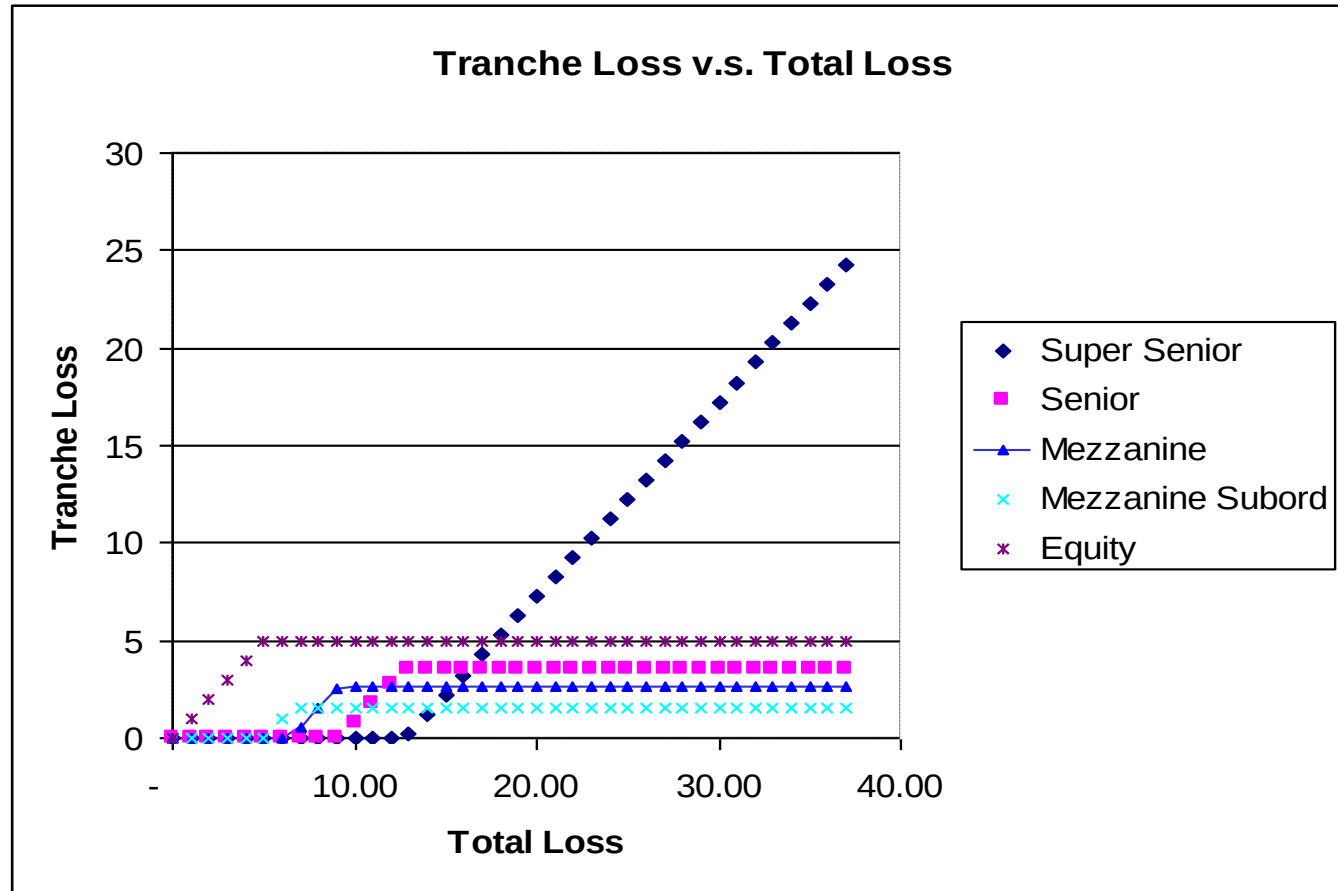
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Tranche Loss as an Option on the Total Portfolio Loss

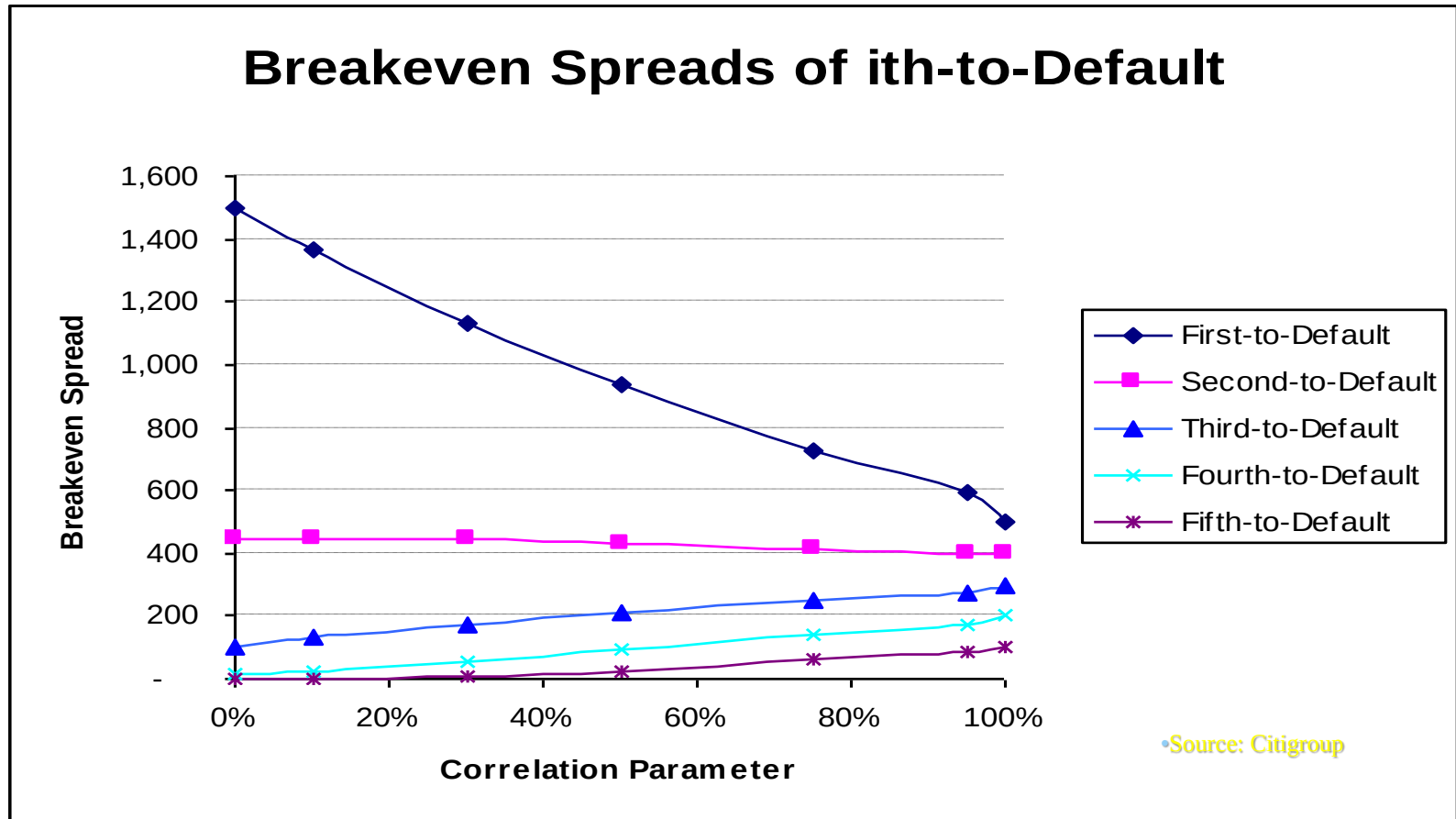


Market Quotes (Nov 30, 2006)

•CDX7		5Y (Ref 34)	delta		7Y (Ref 45)	delta		10Y (Ref 56)
					delta			
•0-3%		23 - 23 1/4	23.0x		40 - 40 1/4	16.0x		51 - 51 1/4
					9.0x			
•3-7%		88 - 89	6.0x		210 - 212	8.5x		428 - 432
					10.0x			
•7-10%		16.5 - 18	1.4x		43 - 45	2.4x		111 - 113
					4.0x			
•10-15%		5.5 - 6.5	0.6x		17.5 - 19	1.1x		51 - 53
					2.0x			
•15-30%		2.5 - 3.5	0.3x		5.5 - 6.5	0.4x		14.5 - 15
					0.6x			

•CDX6		5Y (Ref 31)	delta		7Y (Ref 43)	delta		10Y (Ref 56)
					delta			
•0-3%		19 - 19 1/8	25.0x		37 - 37 1/4	16.5x		51 3/8 - 51 5/8
					9.25x			
•3-7%		59 - 62	4.8x		177 - 182	8.0x		412 - 415
					9.5x			
•7-10%		12.5 - 13.5	1.2x		38 - 40	2.3x		106 - 108
					4.0x			

Correlation and *n*th-to-Default Baskets



Copula Function Framework for Cash CDO Modeling

